Koneru Lakshmaiah Education Foundation
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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Dr Sukanya Ledalla
Student Name	Bhavana Kondakrindi
Roll Number	2520030082
Branch	CSE
Course Outcome	CO4 – Advanced Graph Algorithms & Shortest Paths

Case Study Topic: All-Pairs Shortest Path Analysis using Johnson's Algorithm in Sparse Graphs with Negative Weights.

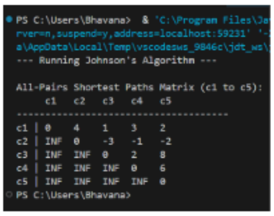
Work Done Summary:

In this case study, we implemented Johnson's Algorithm to solve the All-Pairs Shortest Path (APSP) problem on a directed sparse network containing negative edge weights. The algorithm dynamically checks for negative weight cycles and securely transforms edge configurations to allow optimal, localized search processing.

Implementation Details:

1. Introduced a virtual source vertex q mapped with zero-weight edges to all active nodes in the original network graph.
2. Executed a preliminary Bellman-Ford traversal pass from q to produce potential scaling constants h[] and detect potential negative cycles.
3. Reweighted all network edges into strictly non-negative values using the potential-based algebraic formula: $w'(u, v) = w(u, v) + h[u] - h[v]$.
4. Maintained an edge adjacency structure optimized to execute a sequence of V targeted Dijkstra search cycles running concurrently via a PriorityQueue.
5. Reversed the localized reweighting values upon completion to construct an accurate, unified global shortest-path routing matrix.

Result: Screenshots/output



GitHub Link:

https://github.com/bhavanakondakrindi/DSA-2-PROJECT/blob/main/CO4Assignment.java

Student Signature	Faculty Signature